Instruments and general instrumentation for cavity preparation

Lec. 5

Assis. Prof. Rasha H. Jehad

B.D.S., M.Sc, University of Baghdad College of Dentistry

2025-2026

The removal and shaping of tooth structure are essential aspects of restorative dentistry. Initially this was a difficult process accomplished entirely by the use of

- Hand instruments.

- Rotary, powered cutting instruments.

<u>Hand instruments</u>

Definition:

It is hand-powered dental instruments.

- For many years, ***carbon steel*** was the primary material used in hand instruments because they were harder and maintained sharpness better than stainless steel.
- ***Stainless steels***, which are more corrosion resistant than carbon steels, are now the preferred materials for hand instruments, because all instruments must be sterilized with steam or dry heat between patients and because the properties of stainless steels have improved. There are literally hundreds of formulas for stainless steels, all based on iron and incorporating a significant amount of chromium, some carbon, and very often nickel. Chromium imparts is to improve corrosion resistance and brightness to the metal; carbon imparts improve hardness.
- Hand instruments used in operative dentistry may be categorized as:

1. Cutting instruments and,
2. Non cutting instruments.

Cutting instruments

These instruments are used to cut hard or soft tissue of the mouth. Before rotating instruments were available, dentists could cut well-shaped cavity preparations using sharp hand instruments alone. The process was slow. The advent of the dental handpiece in 1871, first attached to a foot-operated engine, allowed increased speed of tooth preparation. Most tooth preparation today is accomplished with rotary instruments.

Hand cutting instruments are composed of three parts: handle, shank and blade (Fig 1). For non-cutting instruments; the part corresponding to the blade is termed the nib or working end.

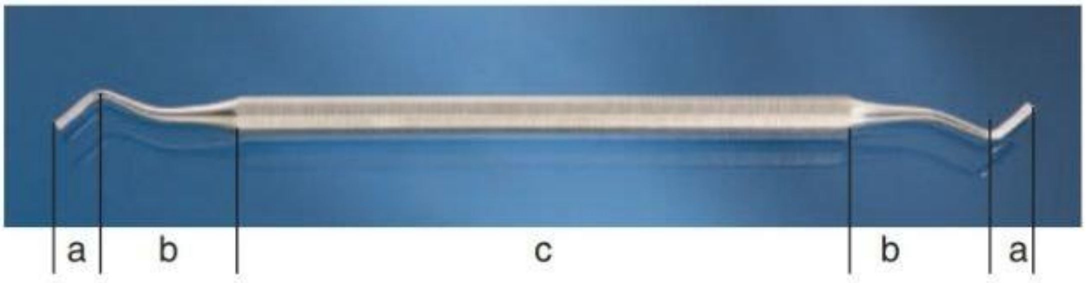
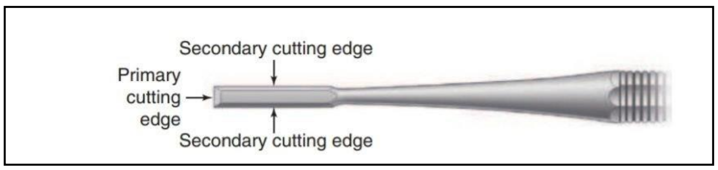


Fig. (1) Double-ended instrument illustrating three component parts of hand instrument: blade (a), shank (b), and handle (c)

- **The blade or nib;** is the working end of the instrument and is connected to the shank.

  The *primary cutting edge* of a cutting instrument is at the end of the blade (called the *working end*), but the sides of the blade are usually beveled and also may be used for cutting tooth structure called the *secondary cutting edge*. Some instrument has a blade on both ends of the handle and are known as double ended instrument. The blades have many designs and sizes, depending on the function they are to perform.



- **The Shank;** serves to connect the handle to the blade of the instrument. The shank may be straight, mono-angle (with one angle), bin-angle (with two angles) , triple-angle (three angles) , or quadr-angle (four angles) .

  The term contra-angle refers to shank in which two or more angles are present.

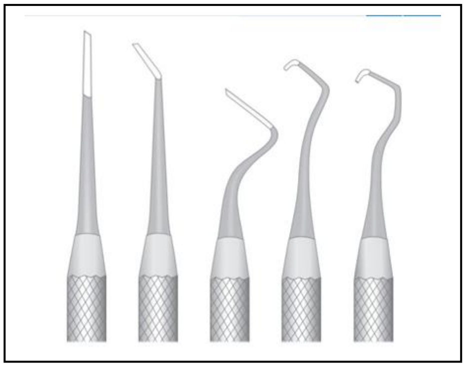
  Balancing of an instrument is achieved by designing the angles of shank so that cutting edge of blade lies within 2–3 mm of long axis of the handle. This principle of instrument design is also called as contra-angling

  Advantages of Balancing of Instruments

  ◆Prevents rotation of instruments when in use

  ◆Generates maximum force at tip of the instrument

  ◆ Improves accessibility and visibility of operating site.



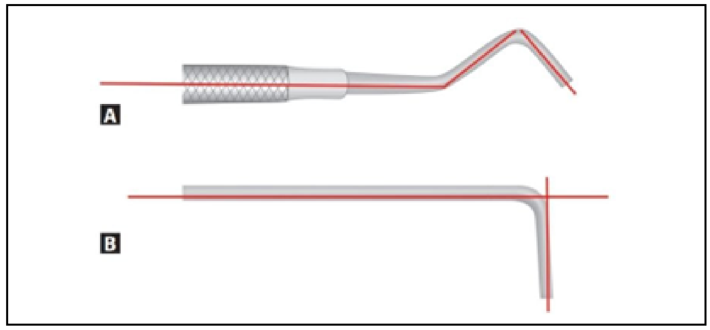
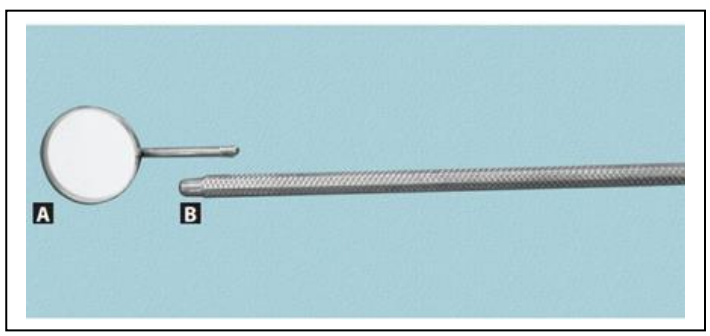


Fig.2 angle of the shank is 2-3mm to handle

The Handle; is the part that is grasped by the operator hand while he is using the instrument. A variety of handle configurations are available:

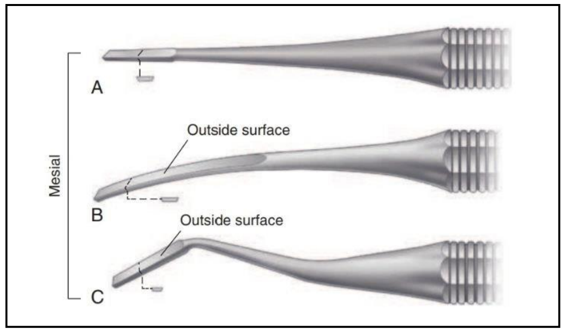
- *Padded handles* are said to increase operator comfort and grip during use.
- *Standard metal handle* has a diameter of approximately ¼ inch (6.4 mm).
- A handle with an intermediate diameter (5/16 inch or 7.9 mm)
- Handles with larger diameters, such as the 3/8-inch (9.5-mm) diameter encouraged primarily for dental hygienists, who spend a large part of their day using hand instruments. A drawback to the use of larger handles in operative dentistry is the space they consume in an instrument tray. Handles can be joined or separable from the shank. Separate type of handle is known as cone socket handle, this allows replacement of various working ends; for example, mouth mirrors and condenser



<u>**Examples of the cutting instruments are:**</u>

Chisel, Hoe, Hatchet, Gingival margin trimmer and excavators

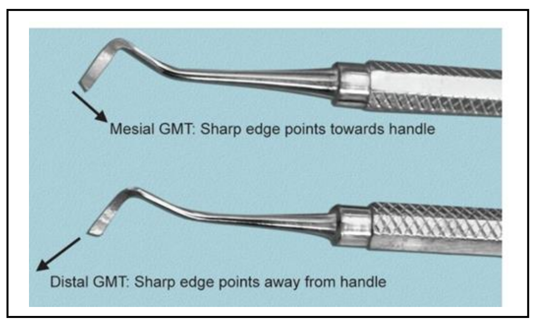
Chisel

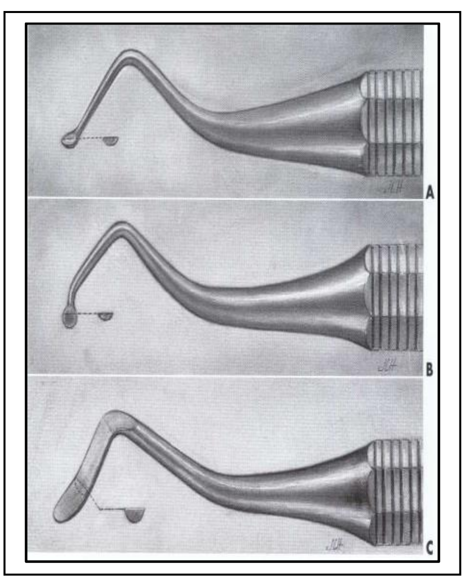


Hoe

Hatchet

Gingival margin trimmer





Excavators

Non-cutting instruments

In these instruments the blade is replaced by a nib or point. These are divided according to **function** into: -

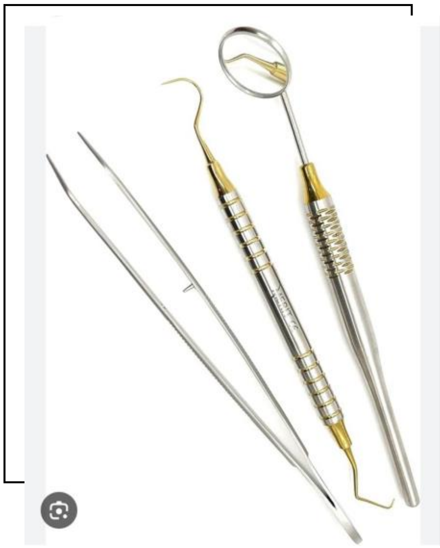
1- Diagnostic instruments

2- Plastic instruments

3- Amalgam instruments

Diagnostic instruments

These are basic instruments that will be needed during each appointment for diagnosis and treatment.



1. **Mirror**: used for:

- Indirect visualization of any tooth surface that cannot be seen by the eye.

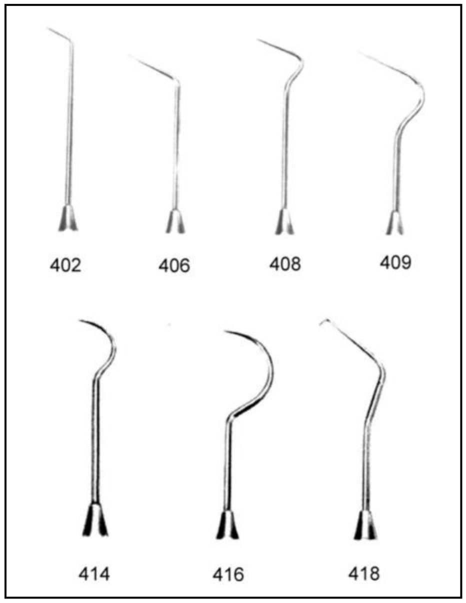
- Reflection of light into the area being examined or treated.

- Retractor of soft tissue (tongue, cheek and lip) to aid access and visualization.

2. **Probe or explorer:** pointed instruments used:

- to feel tooth surface for irregularities
- to determine the hardness of exposed dentin and caries detection.
- Another useful shape is a cowhorn explorer, which provides improved access for exploring proximal surfaces

- Probes have different shapes either sickle, straight or angled



3. **Tweezer or cotton forceps**: used for aiding the operator to carry small items to the mouth of the patient.

Plastic instruments: Or plastic filling instrument are so named because they were originally designed to use with plastic restorative materials, such as acrylic resins, or other nonmetal restoratives, such as the silicates. The name does not refer to the material from which the instrument itself is constructed. They are used to:

- Carry and shape tooth colored restorative material such as composite resin and glass ionomer.
- Packing temporary filling material inside unfilled cavities preparation.
- for placing of basing and lining material into the cavities.

Ash 49: - is double ended instrument with cylindrical nibs and rounded ends. Fig. (9): A

Ash 6: - is one of plastic instrument similar to carver but the margin of its working end is not sharp. Fig. (9): B

Dycal applicator: - is small hand instrument with small round nib used for mixing and placing dycal lining material in the cavity. Fig. (9): C

Cement spatula: - it is used for mixing variety of material which required mixing (such as cement or temporary filling material) on glass or on a paper pad. Fig. (9): D

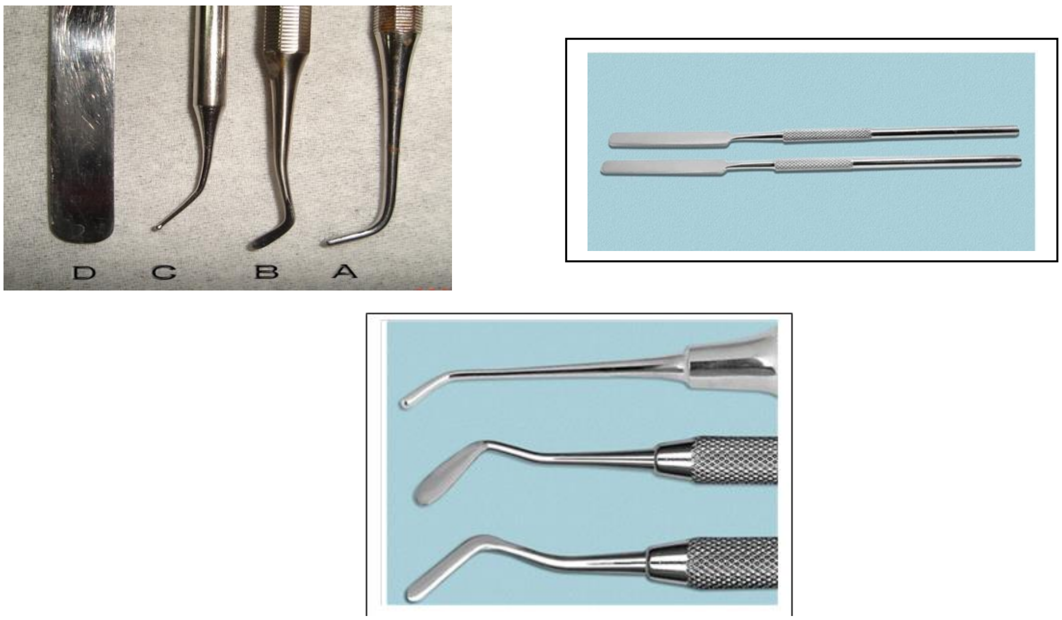
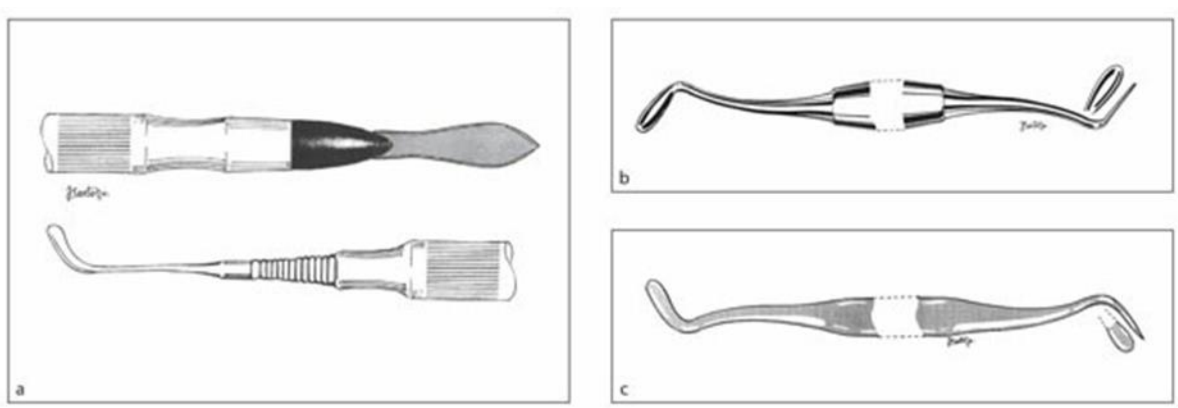


Fig.(9): Plastic instruments

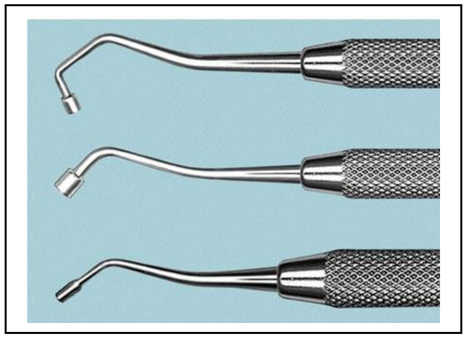
Almore Gold Microfil Instrument, Almore:- A plastic instrument with a large, slightly curved, paddle- shaped blade is very useful for placing and contouring large, anterior resin composite restorations or veneers



Amalgam instruments

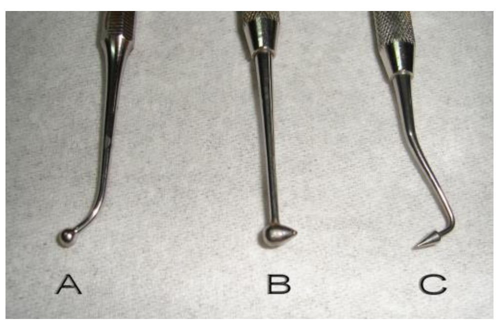
Those instruments used to place dental amalgam, and to a certain extent, resin composite restorative material.

Condensers: - condensers are used to compress the amalgam into all areas of the prepared cavity.

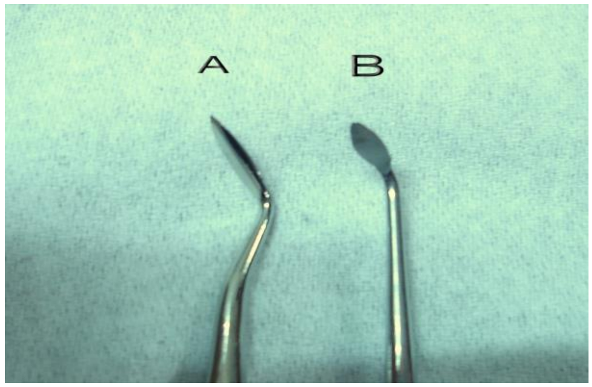


Burnisher: It have different nib shapes: round, oval or rounded cone shapes, also with different sizes, used for several functions such as;

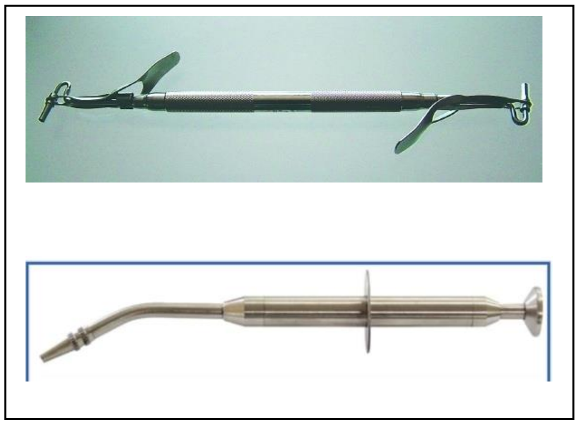
- Shaping metal matrix band to have more desirable contours for restoration.
- sculpting occlusal anatomy in posterior resin composite restorations prior to polymerization of the resin



Carver: - carvers are used to shape amalgam or resin composites (tooth colored) material after they have been placed in the tooth preparations. Carvers have many shapes but usually the nibs are flat with sharp margins for carving. It is important that the blade be sharp.



Amalgam carrier: - used to carry the amalgam and place into the prepared cavities.



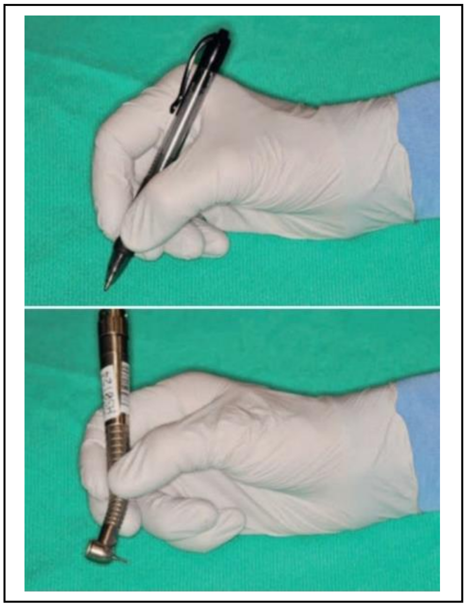
Composite Resin Instruments

These instruments are now available in both hard plastic and metal, and metal instruments are available with several different coatings on blades or nibs to prevent resin materials from sticking to them. The original rationale for using an instrument made of plastic was to eliminate abrasion of metal by the quartz in resin composites, which caused grayness in the tooth-colored material. Because of changes in the inorganic fillers used in many of today's resin composites, the problem of metal abrasion and graying is unusual and material-specific, so even a stainless steel instrument functions well to carry and shape resin composite.

These are set of instruments with a coating of titanium nitride. SinceTitanium Nitride is 40% harder than stainless steel, it is not scratched by filler particles of composite resin. It also resists sticking of resin.

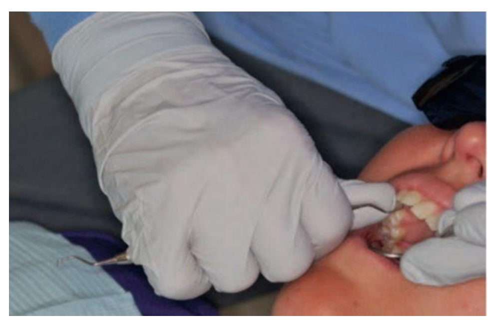
Hand instrument grasping

1. Modified Pen grasp: This is the most frequently used instrument grasp in operative dentistry. The pen grasp is actually different from the way one would grasp a pen; the handle of the instrument is engaged by the end, not the side, of the middle finger; this provides more finger power. The pen grasp is initiated by placement of the instrument handle between the thumb and index finger; the middle finger engages the handle near the shank or on the shank itself. Positioning of the fingers in this manner creates a triangle of forces or tripod effect, which enhances the instrument control .The palm of the operator faces away from the operator .This position stabilizes the instrument and allows the middle finger to help push the instrument down. The ring finger is braced against the teeth to stabilize instrument movement.



2. Inverted pen grasp: - finger positions are the same as for the modified pen grasp except that hand is rotated so that palm faces toward the operator. This grasp is most commonly used for preparing a tooth in the lingual aspect of maxillary anterior and occlusal surface of maxillary posterior teeth with indirect viewing technique

Palm and thumb grasp: - This grasp is same as that for holding the knife for peeling the skin of an apple. Here, instrument is grasped very near to its working end so that thumb can be braced against the teeth so as to provide control during instrument movements. Shaft of the instrument is placed on the palm of the hand and grasped by the four fingers to provide firm control, while the thumb is free to control movements and provide rest on an adjacent tooth of the same arch.



Rotary powered cutting instruments: -

Handpieces: -

Two basic types of handpieces:

- Straight handpiece: the long axis of the bur is the same as the long axis of the handpiece, used for laboratory work.

- Contra angle handpiece: indicates that the head of the handpiece is angled first away from, and then back toward, the long axis of the handle. Also as with hand instruments, this contra-angle design is intended to bring the working point (the head of the bur) to within a few millimeters of the long axis of the handle of the handpiece to provide balance. Used in the mouth.

The contra – angle handpiece are classified according to their speed of rotation into:-

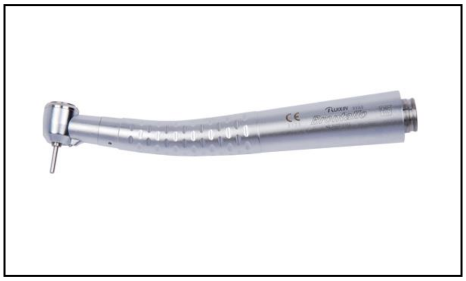
Low speed handpiece: have a free running speed range from 500 to 15,000 rpm (revolution per minute).

- Used for removal of carious dentin with round bur rotating slowly.
- Used with different bur shapes to finish the prepared cavity (e.g. rounding of sharp edges, or flatting of the floor).
- Used in finishing and polishing of restorations.
- The low-speed handpiece will receive only latch-type chuck bur

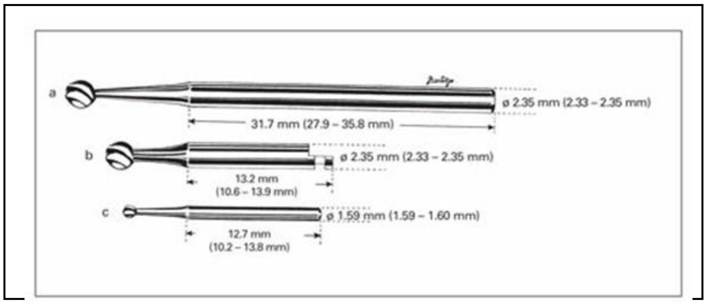
High speed handpiece or air- turbine high-speed handpieces: - have a free running speed above 160,000 rpm and some have speed up to 500,000 rpm. Used with a gentle brushing or painting motion.

- Preferred for cutting enamel and dentin.
- Penetration through enamel and extension of the cavities outline are more efficient at high speed.

- Small diameter burs should be used in the high speed handpiece. High speed generates considerable heat during cutting, even with small diameter burs and should be used with water coolant and high efficiency evacuation.
- The high-speed handpiece will receive only the friction-grip bur



Air- turbine high-speed handpieces



(a) straight handpiece bur; *(b)* latch-type bur for latch-type for slow-speedcontra- angle handpiece; *(c)* friction-grip bur for friction-grip for high-speed contra-angle handpiece.

Burs: -

A group of instruments that can turn on an axis with different speed of rotation to perform different types of work.

- Its work is either cutting, abrasive, finishing or polishing.

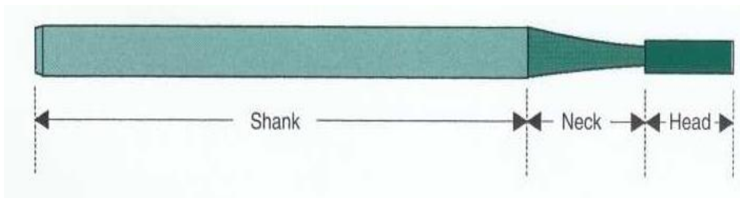
-Burs used for cutting are manufactured from different materials, which may be stainless steel, carbide or diamond.

- The burs have hundreds of shapes and sizes.

Parts of bur:

Shank is the part that fit into the handpiece, accepts the rotation motion from handpiece, and which the bur is locked inside the handpiece head. **Neck** is the part of the bur that connects the head to the shank.

Head is the working part of the bur which contains the cutting edges or points.



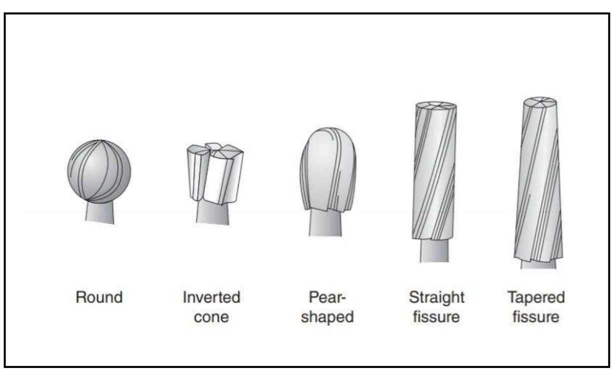
Burs are classified according to the shapes of their heads.

The basic ***bur shapes*** are:

Round bur: the head is spherical so it is used for initial entry into the tooth, preparation of retentive holes or for removal of carious dentin.

Inverted cone bur: the head is a cone – shape with the apex of cone directed toward the bur shank. This bur is used for flatting the floor of the cavity, increasing the depth of cavity or for providing undercuts in cavities preparation.

Fissure bur: is an elongated cylindrical head bur used for obtaining the outline form of the cavity and to cut walls, floor, or margins of the cavity, it's either straight or tapered.



Diamond burs: used with

Cutting in enamel (controlled cutting efficiency so preserve enamel margin).

Finishing the cavity:

Cutting carbide bur: used with

Caries removal: use large bur in removal of deep caries due to controlled heat generation.

Finishing dentin cavity

Diamond burs:

FG Color Code

Blue ring	Medium	106 - 125 µm	ISO 524	M	For removal of enamel
Black ring	Super coarse	180 µm	ISO 544	SC	For rapid gross reduction and removal of old filling
Green ring	Coarse	150 µm	ISO 534	C	For fast reduction of enamel
Green ring	Screw	150 µm	ISO 534	SC	For fast reduction of enamel
Red ring	Fine	40 µm	ISO 514	F	For finishing, crown preparations contouring and finishing composites
Yellow ring	Superfine	20 µm	ISO 504	SF	For fine polishing of composite materials

Finishing and Polishing

Finishing includes the shaping, contouring, and smoothing of the restoration, while ***polishing*** imparts the shine or luster to the surface

Finishing burs:

These finishing burs and diamonds may be used to develop the proper macro- and microanatomy for the restoration.

-The transition from resin to enamel should be slowly smoothed until it is undetectable.

-These burs can be used dry to better visualize the margins and anatomy being developed,

-They should always be used with light pressure to avoid overheating and possibly damaging the resin composite surface.

-***The occlusal surface*** is shaped with a round or oval, 12-bladed carbide finishing bur or finishing diamond.

- Excess composite is removed at the ***proximal margins and embrasures*** with a flame shaped, 12-bladed carbide finishing bur or finishing diamond and abrasive discs.

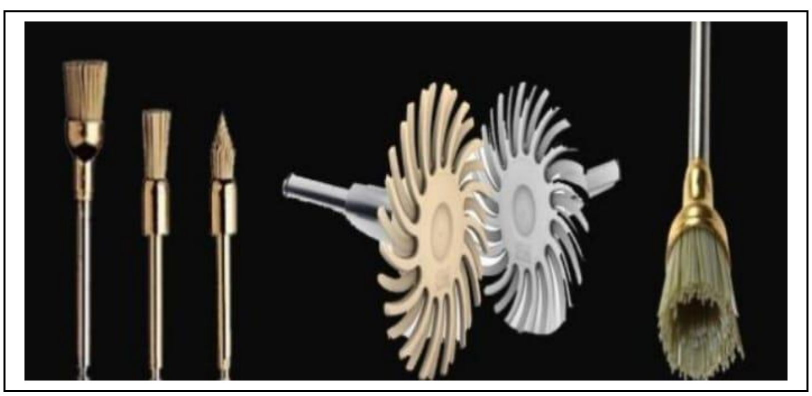
- Any overhangs at ***the gingival area*** are removed with a No. 12 surgical blade

- Narrow finishing strips may be used to smooth the gingival proximal surface.

- To finish ***grooves and the concave lingual surfaces of anterior teeth***, disks, rubber cups and points are used sequentially from coarse to fine grit

Polishing

A final surface polish can be obtained by using a diamond or silicon carbide–impregnated disk or cup. Aluminum oxide or diamond polishing pastes are also used to obtain a high polish.



Guards

Hand instruments like mouth mirror, cheek retractor ,lip retractor, interproximal wedges or clinician’s own finger of other hand are used to protect soft tissues from contact with sharp cutting or abrasive instruments .

These guards are placed in direction of movement of instrument .

Advantages of guards

- Avoid accidental slippage of the instrument
- Prevent tissue injuries

Rubber Dam types and application procedures

Lec. 6

Assis. Prof. Rasha H. Jehad

BDS, MSc, University of Baghdad College of Dentistry

2025-2026

In 1864 S.C. Barnum, a New York City dentist, introduced the rubber dam into dentistry. Use of the rubber dam ensures appropriate dryness of the teeth and improves the quality of clinical restorative dentistry.

Advantages of rubber dam:

Patient related factors:

▪ Provides comfort to the patient.

▪ Protects the patients from swallowing or aspirating foreign bodies.

▪ Protects the patient’s soft tissues by retracting them from the operative field

Operator related factors:

▪ Stress free environment.

▪ Dry, clean operating field.

▪ Magnification is better to be used with rubber dam.

▪ prevents Contamination of restorative material led to decrease in physical properties, also Prevents contamination of tooth preparation (itched enamel and dentin) lead to decrease bond strength.

▪ Moisture control (saliva, sulcular fluid & gingival bleeding).

▪ Avoid any delay.

▪ Infection control by minimizing aerosol production.

▪ Increased accessibility to operative site (access to 2nd molar).

▪ Less fogging of the dental mirror.

Disadvantages of using a rubber dam:

▪ Takes time to apply (but saves more time during procedure)

▪ Cost.

▪ Communication with the patient can be difficult.

▪ Incorrect use may traumatize the gingival tissues.

▪ Insecure clamps can be swallowed or aspirated.

Indication & contraindication:

Rubber Dam is indicated for any case and for every case, except:

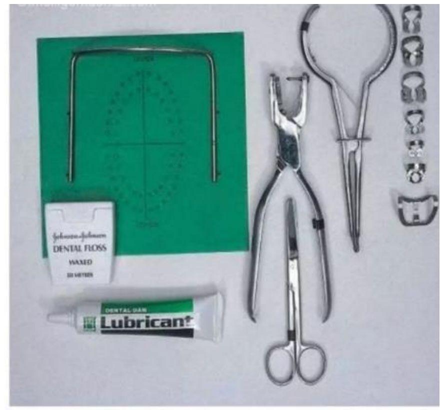
▪ Asthmatic patients.

▪ Epileptic patients.

▪ Mouth breathers.

▪ Extremely malpositioned tooth.

▪ Third molar (in some cases).



Rubber Dam Isolation Kit

Components of rubber dam:

1. Sheets. 2. Template 3. Frame. 4. Forceps. 5. Punch. 6. Clamps. 7. Scissors.

1. Sheets

Colors: ▪ light blue, green, white, purple, black.

light blue: provides: better contrast, more light and Good photos.

Thickness:

Thin (0.15mm) this thickness is indicated in teeth with tight contact areas. It can be used in lower anterior teeth & partially erupted posterior teeth.

Medium (0.2mm) It is indicated in general for all teeth.

Heavy (0.25mm) It has the advantage of providing great adaptation around the teeth & does not tear easily but it exerts much force on the lips & cheek.

Surface: ▪ Dull surface: avoid reflection of light during photographing. ▪ Shiny surface.

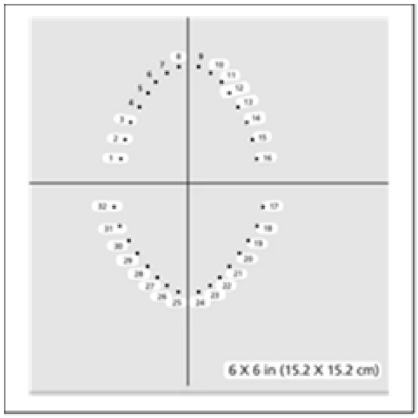
Size:

5 x 5 inch (12.5 x 12.5 cm) usually used for children

6 x 6 inch (15 x 15 cm)

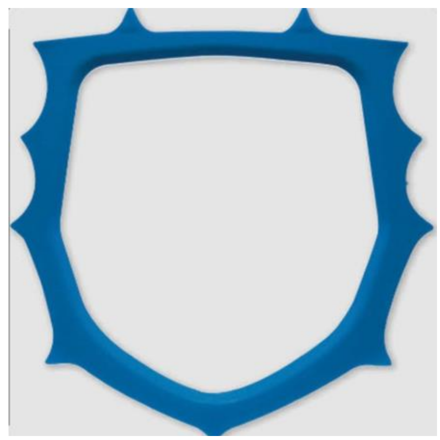
2. Template

Templates are available to guide the marking of the dam. These templates are approximately the same size and shape as the unstretched rubber dam itself. Holes in each template correspond to tooth positions. The template is laid over the dam, and a pen is used to mark through selected holes onto the dam. With the template, the dam can be marked and punched before the patient is seated



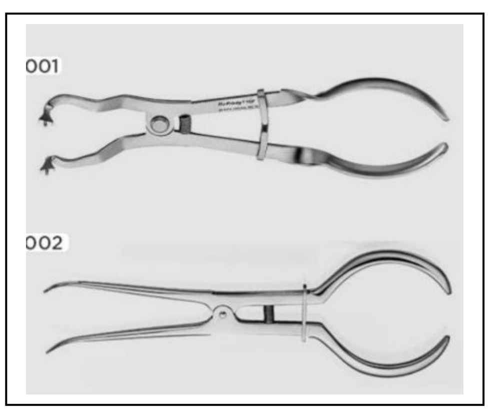
3. Frame

Used to maintain tension in rubber dam. So, it helps in retraction of lip and cheek and tongue. It may be metal or plastic. A plastic frame is advantageous when radiographs will be a part of the procedure because it is radiolucent.



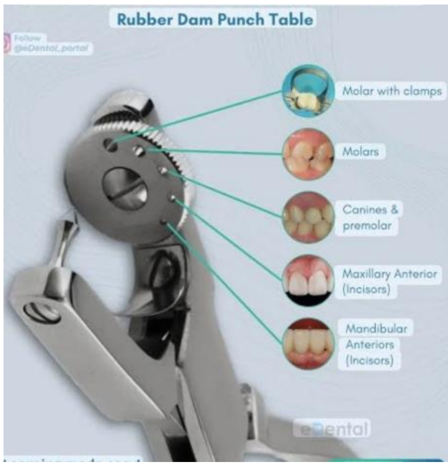
4. Rubber dam holder (clamp forceps)

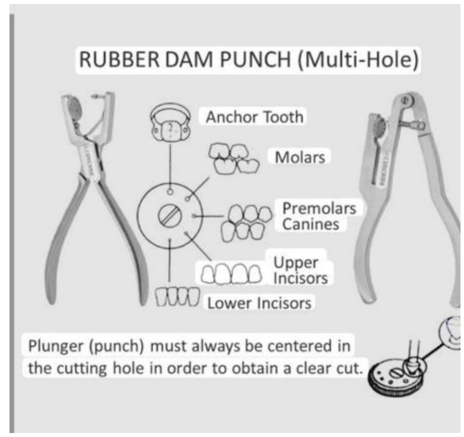
It is used for placement and removal of the retainer from the tooth. There are two designs of forceps: 1- Straight one. 2- S-shape one: it is better than straight one as it is widely opened.



5. Punch

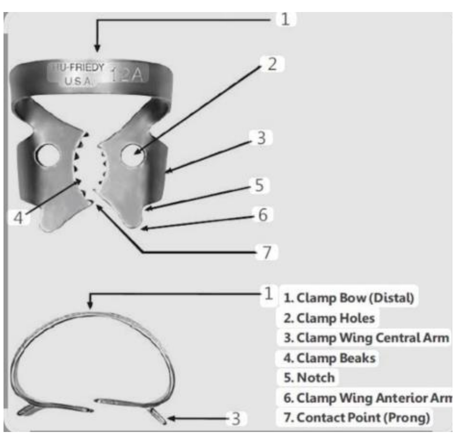
The rubber dam punch is an instrument used to create a hole in rubber dam, having a rotating metal table disc (cutting table) with holes of varying sizes and a tapered, sharp-pointed plunger. The hole should be clear without any tags or tears.

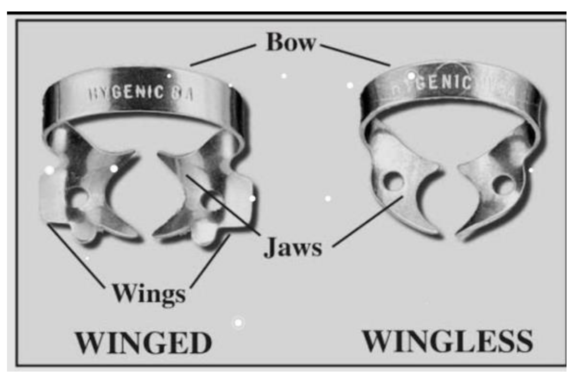




6. Clamps (Retainer)

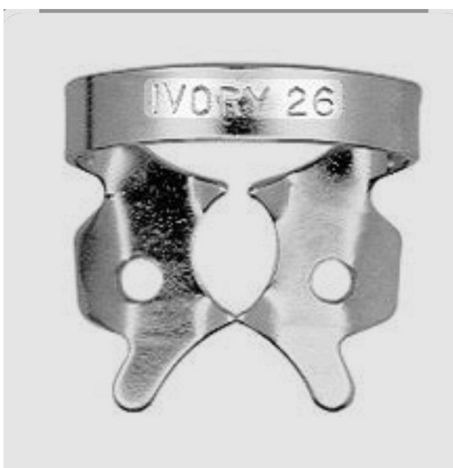
The rubber dam retainer consists of four prongs and two jaws connected by a bow. It is used to Anchor/retain dam on teeth. When positioned on a tooth, a properly selected retainer should contact the tooth on its four contact points (prongs) to prevent rocking or tilting of the retainer. Wingless and winged retainers are available





Ivory clamp for Incisors and bicuspids

Ivory No. 1 for Bicuspids



Ivory No. 26 for Molars

Ivory No. 0 for Incisors and cuspids multiple isolation

7. Scissors

Scissors are often useful in preparing the dam for insertion and are a necessity for cutting the dam for removal.

Methods for application of rubber dam

1. Clamp First Method

*The clamp is placed directly on the tooth first, using forceps.

*After securing the clamp, the rubber dam sheet (with pre-punched hole) is stretched over the clamp and tooth.

*A frame is then placed to secure the dam.

Advantage: Good visibility, especially for beginners.

Disadvantage: More difficult to stretch the dam over the clamp. (Tearing of the dam)

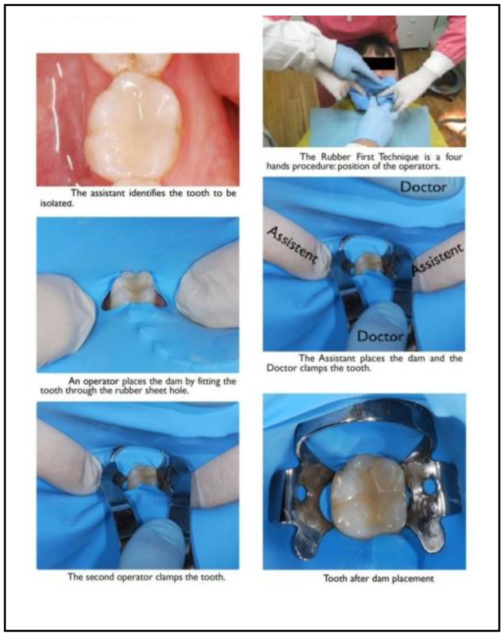
2. Dam First Method

* The rubber dam sheet (with punched hole) is first stretched over the tooth

* Then the clamp is placed through the dam and onto the tooth.

Advantage: Easier for anterior teeth.

Disadvantage: Can be difficult to seat the clamp properly. (Need an assistant help)



3. Clamp and Dam Together

* A winged clamp is placed into the punched hole of the rubber dam.

* Both clamp and dam are carried together to the tooth using forceps.

Advantage: Quick, secure, commonly used in endodontics. (No need for assistant help)

Disadvantage: May obstruct vision during placement.